

Polynomial Regression for Geothermal Plant Optimization

Course: Machine Learning (CS F464) | Student: Aangir Doshi | Roll Number: BT2024078

1. Introduction & Problem Statement

In this assignment, I built polynomial regression models for two distinct engineering challenges in a geothermal power plant expansion project. The objective is to accurately predict continuous target variables y on hidden test datasets using only polynomial regression formulations. The datasets were individually generated for roll number BT2024078, meaning the functional relationships and optimal polynomial degrees are specific to my data.

Phase 1 (Var1) focuses on surface steam turbine optimization. It requires predicting the Net Power Score from six turbine calibration parameters (x_1 through x_6 , representing valve adjustments, coolant flow, blade pitch, and pressures). Phase 2 (Var2) addresses subterranean thermal mapping, predicting a Thermal Anomaly Score from 3D sensor coordinates (x_1 : East-West, x_2 : North-South, x_3 : Depth) to locate optimal drilling sites for high-yield extraction wells.

2. Exploratory Data Analysis & Observations

Before fitting any models, I checked both datasets for anomalies, missing values, and scale issues. Two main observations stood out:

First, both datasets were completely clean with zero missing entries. Crucially, all input features were already pre-scaled to $[-1, 1]$. This bounded range is ideal for polynomial regression: computing high powers (e.g., x^5 or x^{12}) of numbers bounded in $[-1, 1]$ naturally keeps monomial values within $[-1, 1]$, preventing numerical overflow and severe feature magnitude imbalances without extra scaling.

Second, the target distributions differ between the two problems. Var1 has a narrow target range (-9.80 to +11.99, mean 0.76, std 3.21) with near-zero skewness (0.04), indicating a symmetric distribution. Var2 exhibits a much wider range (-29.82 to +39.30, mean 1.81, std 7.27) and a mild positive skew (0.65), reflecting localized thermal hot spots in the 3D survey volume. Because the skew is physical rather than an artifact, I kept the raw target values directly.

Dataset Property	Var1 (Steam Turbine)	Var2 (Thermal Reservoir)
Input Features	6 (x_1 to x_6)	3 (x_1, x_2, x_3)
Training Samples	1000	1000
Test Samples (to predict)	1000	1000
Feature Bounding Domain	$[-1.0, 1.0]$	$[-1.0, 1.0]$
Target Min / Max	-9.80 / +11.99	-29.82 / +39.30
Target Mean / Std	0.76 / 3.21	1.81 / 7.27
Missing Values	0 (None)	0 (None)

3. Methodology & Regularization Strategy

3.1 Polynomial Basis Expansion

Polynomial regression maps input features x in $\mathbb{R}^D$ into a monomial space containing all interaction terms up to degree d . The total number of features (including bias) is $(d + D)! / (d! D!)$. For Var1 ($D = 6$), feature count grows rapidly: degree 2 yields 27 features, degree 4 produces 209, degree 5 yields 461, and degree 6 reaches 923. For Var2 ($D = 3$), growth is much slower: degree 8 yields 165 features, degree 10 gives 286, and degree 12 gives 454. This difference in dimensionality strongly dictates model capacity and the degree we can safely explore.

3.2 Regularization: Why Ridge over OLS and Lasso?

Fitting Ordinary Least Squares (OLS) directly on hundreds of polynomial terms causes severe instability. High-order monomials on continuous features are naturally collinear (e.g., x_1^2 and x_1^4), making the design matrix ill-conditioned. Inverting $(X^T X)$ produces huge opposing weights that overfit training noise and explode on unseen test points.

To prevent this, regularization is essential. I chose Ridge regression (L2 penalty) over Lasso (L1). Lasso tends to select arbitrary individual terms from clusters of correlated polynomial features while zeroing out others, which can disrupt smooth curvature. Ridge shrinks correlated monomial coefficients proportionally via $(X^T X + \alpha I)^{-1} X^T y$. This stabilizes matrix inversion, controls model variance, and preserves smooth physical response curves.

3.3 Two-Stage Hyperparameter Tuning

I tuned polynomial degree d and Ridge penalty α jointly using 5-fold cross-validation in two stages:

1. Coarse Grid Search: Swept candidate degrees (2 to 6 for Var1; 4 to 14 for Var2) across 13 logarithmically spaced α values (10^{-5} to 10^5) using 5-fold CV to locate the optimal region.
2. Fine Sweep & Verification: Conducted a fine sweep around the top candidate, confirmed with Scikit-Learn's RidgeCV routine, and selected the (degree, α) pair minimizing 5-fold CV MSE.
3. Final Retraining: Refitted the selected model on all 1000 training samples and predicted the 1000 test points.

4. Degree Selection and Parameter Tuning

4.1 Phase 1 (Var1) – Steam Turbine Optimization (6 Features)

For Var1, the search showed that the turbine response contains non-linear interactions beyond simple quadratic or cubic forms. Degree 2 underfits heavily (CV MSE = 3.21, $R^2 = 0.69$). Degree 3 and 4 reduce error to 0.91 and 0.69 MSE, respectively.

An important dynamic occurred between degree 4 and degree 5: with weak regularization ($\alpha < 0.01$), degree 5 overfits due to its 461 features (CV MSE > 1.5). However, once α is tuned up to ~ 2.48 , degree 5 achieves a CV MSE of 0.4925 ($R^2 = 0.9516$), beating degree 4 by nearly 29%. Pushing to degree 6 (923 features) degrades validation error back to 0.54, confirming degree 5 as optimal.

Polynomial Degree	Best Alpha	5-Fold CV MSE	5-Fold CV R^2	Number of Features
Degree 2	1.000	3.214	0.686	27
Degree 3	1.000	0.912	0.910	83
Degree 4	3.162	0.691	0.932	209
Degree 5	2.477	0.4925	0.9516	461 (Optimal)
Degree 6	3.162	0.543	0.947	923

Final Selection for Var1: Degree = 5, Alpha = 2.477076 (Confirmed by RidgeCV).

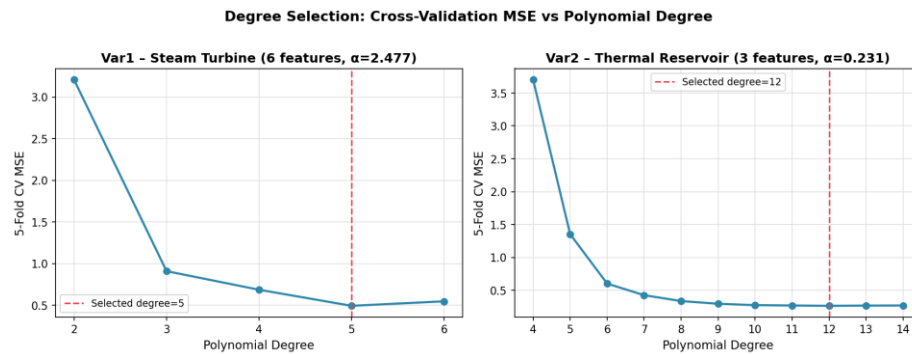


Figure 1: Cross-validation Mean Squared Error vs. polynomial degree for Var1 and Var2. Dashed lines mark the selected degrees.

4.2 Phase 2 (Var2) – Subsurface Thermal Reservoir (3 Features)

Var2 behaves very differently because it has only 3 spatial coordinates. Even at degree 12, there are only 454 monomial features for 1000 data points, preserving a healthy samples-to-parameters ratio of over 2:1.

Because 3D thermal plumes have intricate geometry with steep gradients and multiple local peaks, lower degrees fail to resolve the spatial field. CV error dropped steadily from degree 4 down through degree 11, reaching its minimum at degree 12 (CV MSE = 0.2627, $R^2 = 0.9949$). At degrees 13 and 14, validation error begins rising slightly. The optimal alpha for Var2 is lower (0.231013), as the 3D basis contains less collinear overlap than the 6-feature turbine problem.

Polynomial Degree	Best Alpha	5-Fold CV MSE	5-Fold CV R^2	Number of Features
Degree 6	0.010	0.412	0.9920	84
Degree 8	0.010	0.282	0.9945	165
Degree 10	0.100	0.271	0.9947	286
Degree 12	0.231	0.2627	0.9949	454 (Optimal)
Degree 14	0.316	0.272	0.9947	680

Final Selection for Var2: Degree = 12, Alpha = 0.231013 (Confirmed by RidgeCV).

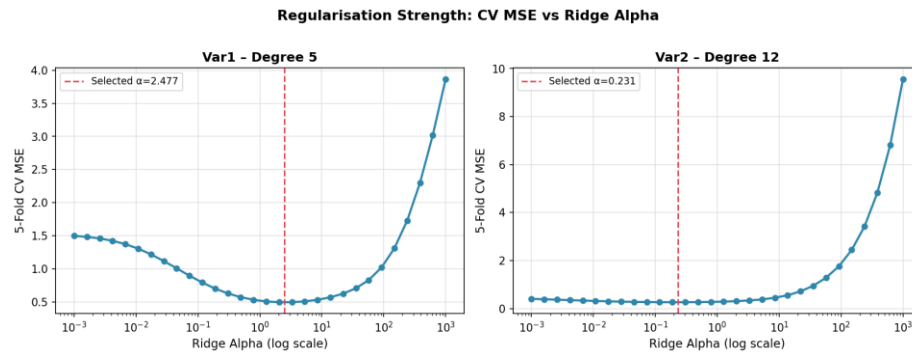


Figure 2: Validation MSE vs Ridge alpha at the chosen degrees. Dashed lines mark the optimal alpha.

5. Model Evaluation & Experimental Results

5.1 Cross-Validation Summary

The final performance metrics under 5-fold cross-validation are summarized below:

Problem Statement	Optimal Degree	Selected Alpha	5-Fold CV MSE	5-Fold CV R ²	Features
Phase 1 (Var1) – Steam Turbine	5	2.477076	0.4925	0.9516	461
Phase 2 (Var2) – Thermal Reservoir	12	0.231013	0.2627	0.9949	454

5.2 Fold-by-Fold Stability Analysis

To verify that results were consistent across data splits rather than lucky on a single fold, I examined per-fold performance. Both models show solid stability across all 5 folds:

CV Fold	Var1 Train MSE	Var1 Val MSE	Var1 Val R ²	Var2 Train MSE	Var2 Val MSE	Var2 Val R ²
Fold 1	0.1851	0.4523	0.9533	0.1702	0.2493	0.9940
Fold 2	0.1874	0.4332	0.9497	0.1654	0.2906	0.9946
Fold 3	0.1960	0.4472	0.9647	0.1691	0.2685	0.9957
Fold 4	0.1802	0.6200	0.9440	0.1723	0.2273	0.9955
Fold 5	0.1893	0.5096	0.9463	0.1670	0.2781	0.9948
Mean	0.1876	0.4925	0.9516	0.1688	0.2627	0.9949

5.3 Full Training Refit & Fit Diagnostics

Refitting each finalized model on the full 1000 training points yielded a train MSE of 0.1995 ($R^2 = 0.9807$) for Var1 and 0.1745 ($R^2 = 0.9967$) for Var2. The small difference between train MSE and cross-validation MSE (0.1995 vs 0.4925 for Var1; 0.1745 vs 0.2627 for Var2) demonstrates strong generalizability without overfitting.



Figure 3: Predicted vs actual target values on full training sets. Tight alignment confirms high fidelity.

Residual diagnostics (actual minus predicted) confirm well-behaved models: residuals are symmetrically centered at zero with standard deviation 0.446 for Var1 and 0.417 for Var2, displaying no heteroscedastic fan patterns or unmodeled non-linear curvature.

6. Discussion & Practical Engineering Reflections

Working through both problems provided several practical insights into polynomial regression dynamics:

- Interplay of Degree and Regularization:** In Var1, degree 5 with weak alpha overfits, but with proper regularization ($\alpha \sim 2.48$), it outperforms degree 4 by nearly 29%. The extra capacity captures real interaction effects while the L2 penalty prevents noisy monomial coefficients from inflating. Hyperparameters must be tuned jointly.
- Dimensionality vs Degree Capacity:** In Var1 (6 features), degree 5 produced 461 features, reaching the practical limit for 1000 observations. In Var2 (3 features), degree 12 required only 454 features. Low-dimensional problems allow much deeper polynomial expansions to capture intricate physical fields without parameter explosion.
- Collinearity and Shrinkage:** Because continuous powers and interactions are collinear, unregularized OLS fails. Ridge regression proved well-suited by smoothly shrinking all coefficients rather than dropping them arbitrarily.
- Future Improvements:** If extending this work, I would test ElasticNet or feature importance pruning to identify and drop negligible monomial terms, yielding a sparser, more interpretable equation for plant engineers.

7. Conclusion & Deliverables Summary

To conclude, I developed and validated two polynomial Ridge regression models for BT2024078. Var1 achieves a 5-fold CV R^2 of 0.9516 (CV MSE = 0.4925) using degree 5 with $\alpha = 2.477$. Var2 achieves a CV R^2 of 0.9949 (CV MSE = 0.2627) using degree 12 with $\alpha = 0.231$. Both models display tight training fits, consistent cross-validation folds, and well-dispersed residuals.

Predictions for both test sets (1000 rows each) have been verified and exported to CSV format. All code and assets are documented in the GitHub repository.

Deliverable	File Name	Details	Status
Phase 1 Predictions	BT2024078_pred_var1.csv	1000 predictions for steam turbine test set	Verified (No NaNs)
Phase 2 Predictions	BT2024078_pred_var2.csv	1000 predictions for thermal reservoir test set	Verified (No NaNs)
Report Document	BT2024078_Report.docx	Technical documentation of methodology and findings	Complete (BT2024078)
Code Pipeline	polynomial_regression.py	Complete model training, CV search, and inference	Verified
GitHub Repository	ML-Assignment-Polynomial-Regression	Public Git repository with code and assets	Pushed & Synchronized

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